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Late Palaeozoic palaeomagnetic data for Wrangellia: resolution of the polarity ambiguity

Bruce C. Panuska & David B. Stone

Geophysical Institute, University of Alaska, Fairbanks, Alaska 99701, USA

Wrangellia and the Alaska Peninsula terrane are two of the many allocthonous terranes making up southern Alaska today. Palaeomagnetic data from these areas clearly indicate low palaeolatitudes in the early Mesozoic. Because of local and regional rotations, the mean declination of the palaeomagnetic vectors is uncertain. This, in addition to incomplete temporal coverage, has led to an ambiguity in the polarity of the vector, and hence an ambiguity in whether the palaeolatitudes determined are in the Northern or Southern Hemisphere. The data considered here have resolved the ambiguity for Wrangellia, and by implication for the Alaska Pennisula terrane, and also allow the construction of a highly speculative apparent polar wander curve.

A major portion of the North American Cordillera is composed of 'suspect' terranes[1,2]. These tectonostratigraphic terranes are suspected of being allochthonous to cratonic North America. Palaeomagnetic studies have shown that many of these terranes have travelled great distances[3]; however, the geographic origin and tectonic history, in terms of plate motion, are still very poorly understood.

Palaeomagnetic data, particularly from severely deformed areas, are often difficult to interpret for use in palaeogeographical reconstructions. A palaeomagnetic pole may yield an estimate of the palaeolatitude and the amount of rotation necessary to reconcile any discordance between the ancient geomagnetic pole position and the present geographical pole, but gives no longitudinal control. There is also a polarity ambiguity which is particularly important at a low palaeolatitudes. As the Earth's magnetic field reverses polarity from time to time, it is not apparent whether the observed north seeking magnetic vectors correspond to the north geographical pole (normal polarity) or the south geographical pole (reversed polarity). The polarity will affect the sign of the palaeolatitude (Northern or Southern Hemisphere) and the amount of relative rotation needed to bring the palaeomagnetic pole into coincidence with the reference pole. Due to the sparse data base, especially with regard to time, the palaeomagnetic signatures of the terranes in Alaska are only poorly known; there are thus few guidelines to aid in the determination of the correct polarity. Consequently, various workers have assigned polarities to minimize the required amounts of rotation, give continuity of motion and give reasonable plate velocities. These arguments are rarely convincing and the polarity dilemma has hampered construction of polar wander curves. There are two techniques to resolve the ambiguity of polarity. One is to construct a continuous apparent polar wander curve backwards in time from the present. This has not yet been possible for southern Alaska because of the difficulty of obtaining samples which have a wide enough distribution in time from appropriate locations. The construction of an apparent polar wander curve is also hampered by lack of precision in relating the measurements of the ancient declination of the geomagnetic field from one area to another. This difficulty arises from the fact that relative rotations of individual terranes may have occurred, as are seen elsewhere in Western North America[3], and apparent local rotations may also arise through multiple folding events.

The second technique, which is applied here, is to use data from times when the geomagnetic field was anomalously stable. One of these times, the Permo–Carboniferous reversed magnetic interval, the geomagnetic field was almost exclusively of one polarity (reversed), times of normal polarity being both very short and extremely rare. The data discussed here are from rocks formed at the beginning of this period (when short normal polarity intervals were slightly more frequent) and in the middle (when they are extremely rare), thus allowing great confidence in our assumption of reversed polarity. We now report late Palaeozoic palaeomagnetic data of known polarity for the Wrangellia terrane[4] and explore some of the ramifications for polar wander curves and tectonic history.

As originally described, the Wrangellia terrane[4] comprises several tectonic units recognized in the Wrangell Mountains, Alaska, parts of southeastern Alaska, Queen Charlotte Island, Vancouver Island and an isolated fragment in eastern Oregon. Hillhouse[5] presented palaeomagnetic data from Triassic rocks in the Wrangell Mountains which document low palaeolatitude, within 15° of the Equator. This low Triassic palaeolatitude has now been confirmed for other localities within Wrangellia[6–9]. Hillhouse[5] assigned normal polarity to the pole located at 146° E, 2° N on the basis that it is closer than the antipole to the equivalent Triassic pole for cratonic North America, and requires the lesser amount of northward translation and rotation for Wrangellia.

Three new localities have been sampled along Skolai Creek in the Wrangell Mountains, Alaska. These sample localities lie close to and stratigraphically beneath Hillhouse's[5] Golden Horn Triassic locality (Fig. 1) and form part of the same, apparently intact stratigraphic sequence. The present results may thus be directly related to the Wrangellian Triassic palaeomagnetic poles, as these localities all seem to have experienced the same overall deformational history. Taking regional attitudes into account they may also be compared with cratonic North America.

Seventeen oriented cores were collected from both the Lower Permian Hasen Creek Formation (volcaniclastic sediments) and the Lower Permian or Pennsylvanian Station Creek Formation (volcanic, pyroclastic and volcaniclastic sedimentary rocks). In addition, six cores were obtained from a gabbro (Middle or Upper Triassic) intruding both the Hasen Creek and Station Creek Formations.

All samples were subjected to stepwise thermal demagnetization to remove secondary magnetic components. Primary vector directions were determined for each core on the basis of mininum change of direction over successive demagnetization steps, accompanied by systematic intensity decreases. Samples that did not display a stable vector direction for at least the last demagnetization steps were not used for calculation of a mean vector direction. In addition, vector directions more than an arbitrary 40° from the mean were eliminated from the calculation of the final mean on the basis that they probably represent either excursions of the geomagnetic field or errors incurred during the collection and processing of samples. Only

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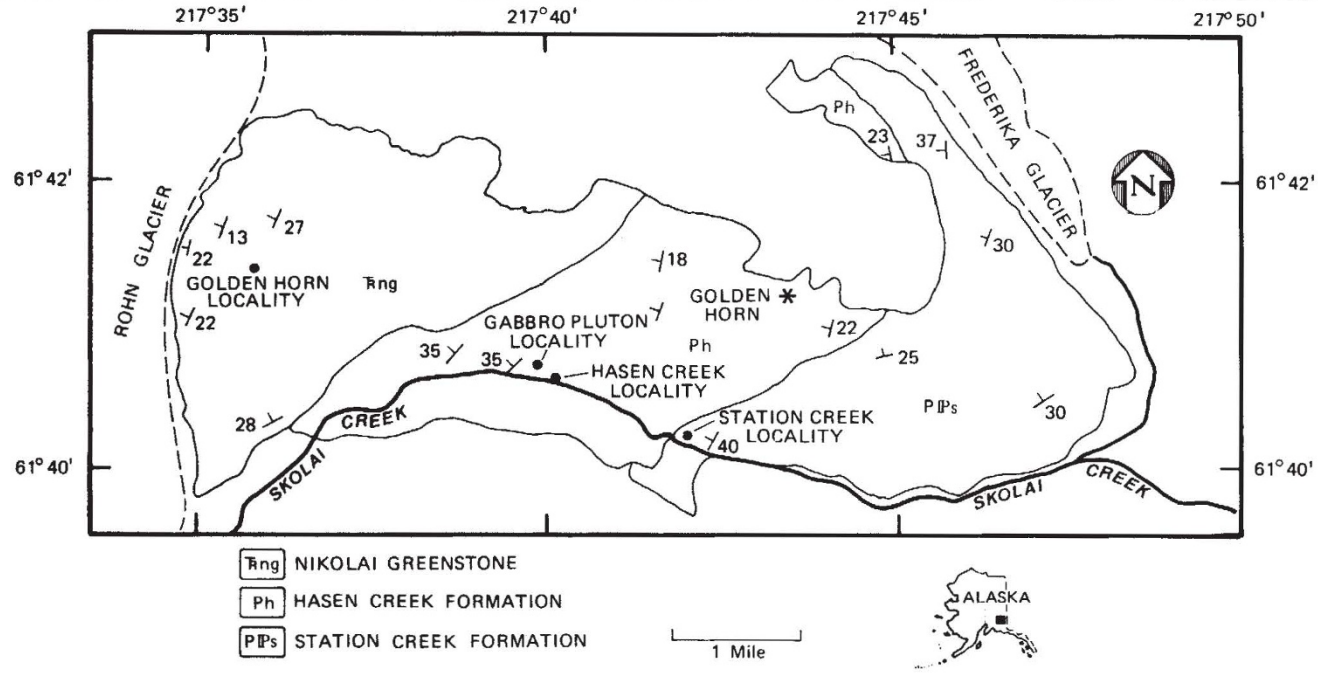


Fig. 1 Generalized geological map of Skolai Creek valley, eastern Wrangell Mountains, Alaska. Modified from ref. 16.

four vectors were rejected for these reasons (all from the Station Creek Formation). For all three localities, the mean vector directions with respect to present geographical coordinates do not include the present field within their circles of confidence. The fold test for these samples is positive, inasmuch as the precision parameter (k) (ref. 10) is larger when the data are calculated with respect to local indications of ancient horizontal than with respect to present horizontal. In addition, the Station Creek Formation suite contains one vector direction whose polarity is the opposite of the others. The mean directions (Table 1) thus represent a reliable measure of the ancient geomagnetic field.

The polarity for the virtual geomagnetic poles (VGP) can be assigned with confidence, as the Hasen Creek Formation is known from reliable fossil evidence to be of Permian age[11], which corresponds to the Permo–Carboniferous reversed magnetic interval[12]. While there are rare normal magnetic intervals in this 'quiet' period, the probability of sampling only a normal interval in a thick (40 m) stratigraphic sequence is exceedingly small. Also, the existence of a single reversal in the earlier Pennsylvanian–Permian Station Creek suite is consistent with deposition within the section of the Permo–Carboniferous quiet interval which had a few short normal polarity episodes. Thus, taking into account the reversed geomagnetic polarity, the VGPs corresponding to the mean vectors plotted in Fig. 2 represent an estimate of the location of the north geographical pole for that time.

The VGPs for the Hasen Creek and Station Creek localities plot very close together. The VGP for the pluton is displaced towards the Triassic Wrangellian pole of Hillhouse[5]. Because a direct measure of stratigraphic tilt was not possible for the pluton, the tilt correction had to be assigned on the basis of local geology, and is thus inherently less accurate than tilt corrections for the layered rocks. However, the general positon of the VGP for the pluton supports the validity of the other VGPs.

The VGP determinations reported here for the Pennsylvanian–Permian are located within 30° of the Triassic VGPs described by Hillhouse[5]. This provides strong evidence that Hillhouse's polarity preference is correct and that Wrangellia was located at lat. ~15° N, during late Pennsylvanian (?) to Triassic time.

Jones and Silberling[13] suggest that Wrangellia and the Peninsular terrane had amalgamated by at least late Jurassic time, based on biostratigraphic considerations. If this interpretation is correct, then existing palaeomagnetic data for the Peninsular terrane should be applicable to Wrangellia from Jurassic time to the present. There is, however, a polarity ambiguity for the Jurassic Peninsular pole.

A Southern Hemisphere polarity for the Jurassic Peninsular pole has been proposed[7,8] on the basis of 'smoothest' motion and minimum number of changes in direction of the inferred plate motion. The confirmed Triassic pole for Wrangellia is considerably closer to the Southern Hemisphere Jurassic pole for the Peninsular terrane than the Northern Hemisphere pole. Selec-

Table 1 Magnetic field data from formations in southern Alaska

Formation	*Dg*	*Ig*	*k*	*Ds*	*Is*	*k*	α95	Long. °N	VGP Lat.	*k*	α95	*N*
Hasen Creek (Lower Permian)	207.8	14.0	21.0	218.0	14.5	22.2	9.0	178.6	−14.9	41.3	6.6	11
Gabbro pluton	—	—	—	242.0	−2.8	111.0	5.4	152.1	−14.2	192.1	4.1	6
Station Creek (Lower Permian–Pennsylvanian)	215.4	1.8	3.1	225.9	10.9	8.9	14.9	167.6	−13.5	14.4	11.7	10

Dg, *Ig* are the declination and inclination of the magnetic vector with respect to geographical coordinates (present north and present horziontal); *Ds*, *Is* are the declination and inclination of the magnetic vector with respect to a stratigraphic frame of reference (present north and ancient horizontal); VGP is a virtual geomagnetic pole reported in terms of geographical longitude and latitude: k is Fisher's precision parameter; N is the number of samples. Geographical vectors for the pluton were not reported as all samples were assigned the same tilt correction; thus a fold test was not possible. The polarity of the tabulated data represent the equivalent of today's Northern Hemisphere pole.

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tion of the Northern Hemisphere Jurassic pole as the correct one would also require ~125° of rotation. Similar rotations have been documented for numerous terranes[3]: however, these are considered to be related to regional shear in a transform setting. Evidence for similar transform tectonics for Wrangellia in Triassic or Jurassic time is lacking. On this basis the Southern Hemisphere Jurassic pole is favoured, which then allows a highly speculative polar wander curve to be drawn (Fig. 3) through the above Pennsylvanian–Permian data, the Triassic VGP of Hillhouse[5] and the Jurassic, Cretaceous and Eocene VGPs of Stone[14] and Stone and Packer[15].

Because of possible significant errors in the declination data, due to unknown amounts of relative rotations of the two major terranes sampled, the highly speculative nature of the polar wander curve must be emphasized. Even taking this into account, the apparent polar wander path shows two pronounced 'kinks'; one in Jurassic time and one some time between late Cretaceous and Eocene time. The possibility that these kinks represent changes in the geological regime is enhanced by the fact that the Jurassic aged 'kink' also reflects a change in the trend of the palaeolatitude versus time of the sampling sites. Because the palaeolatitude depends solely on the inclination of the magnetic vector with respect to ancient horizontal, it is unaffected by relative rotation. The timing of these sudden changes of trend in tectonic motion with respect to the geomagnetic, and by implication the geographical pole, are reasonably close in age to two established orogenic episodes in Alaskan geology. We speculate that the Jurassic orogenic event marks the amalgamation of Wrangellia and other southern Alaska terranes, probably including the Peninsular terrane. The later event marks the accretion of this superterrane to North America some time during late Cretaceous to Eocene time. In this model, the kinks in the polar wander path would be the palaeomagnetic signatures of the collision events, which in turn caused readjustments of the motions of the terranes.

If this polar wander path is correct, it also implies that Wrangellia, and whatever plate it was part of, had significant southerly motion during earliest Mesozoic times. This was then followed by northerly plate motion beginning in the Jurassic period which carried Wrangellia and the Peninsular terrane from the Southern Hemisphere to their present location in southern Alaska.

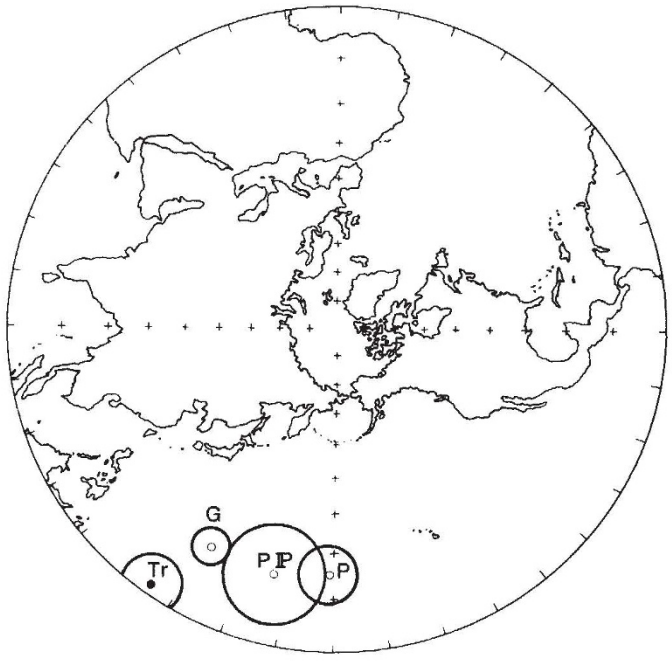


Fig. 2 Virtual geomagnetic pole (VGP) positions and associated circles of 95% confidence for upper Palaeozoic units. P, Hasen Creek Formation (Permian); PP, Station Creek Formation (Permian–Pennsylvanian); G, a gabbro pluton (Triassic). ●, Northern Hemisphere; ○, Southern Hemisphere.

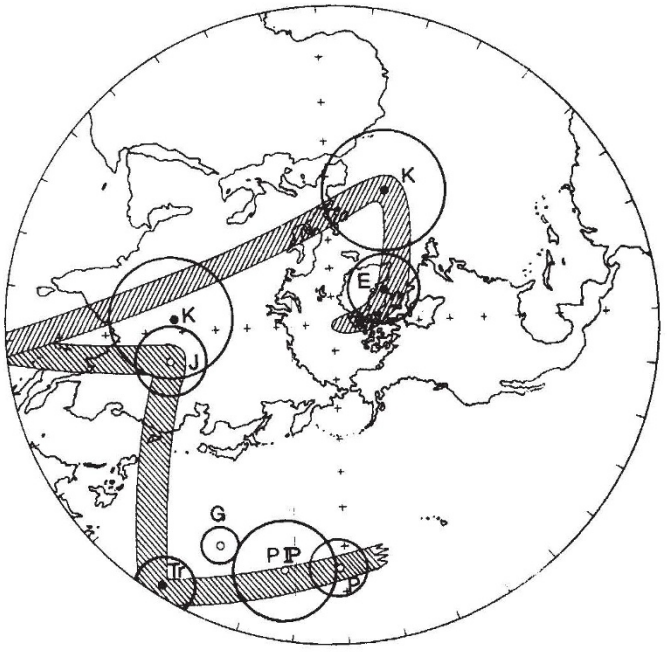


Fig. 3 Speculative polar wander curve for Wrangellia. Permian–Pennsylvanian and Triassic poles are considered to be fixed, inasmuch as the choice of polarity is almost certainly correct. Jurassic to Eocene poles are from the Peninsular terrane, based on the assumption that the Peninsular terrane and Wrangellia were joined by late Jurassic. ●, Northern Hemisphere; ○, Southern Hemisphere.

Obviously we cannot calculate actual rates of plate motion from the data available, but the data do not require excessive motions. The maximum plate velocity needed is ~6 cm yr^{-1} of latitudinal movement between the Jurassic and the present. In principle, it may be possible to make generalized reconstructions of the motions of the Palaeo-Pacific oceanic plate(s) by using allochthonous terranes as 'tracers'. This is an important possibility, as there is no Pacific oceanic crust older than Jurassic from which plate motions can be ascertained directly.

Although more sample localities are desirable to establish firmly the position of the Pennsylvanian–Permian poles, the evidence confirms the polarity of Hillhouse's[5] Triassic palaeomagnetic pole for Wrangellia. This resolves the polarity dilemma for part of the Wrangellia polar wander curve, and will reduce polarity problems in future studies.

The highly speculative polar wander curve for Wrangellia, based on data from both Wrangellia and the adjacent Peninsular terrane, can also be verified (or refuted) by more sampling, particularly in the time-stratigraphic intervals in Wrangellia which currently have sparse or marginal palaeomagnetic data.

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